STRATEGY OVERVIEW · NQ / MNQ

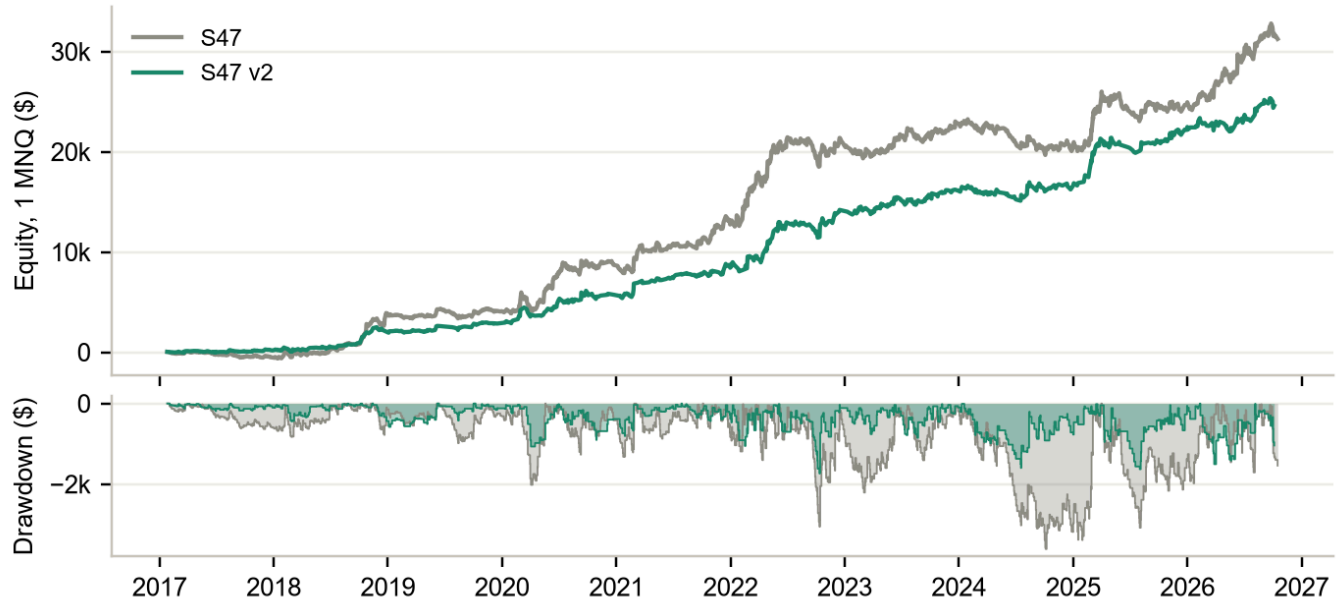
S47 — Europe-Open Impulse, and S47 v2

E-mini Nasdaq-100 · 1-minute bars · Chicago time · one trade a day, long or short · enters 09:31, flat at 15:00 CT · v2 = S47A, "London and New York agree"

1 MNQ, whole backtest	Net	PF	Trades / yr	Win rate	Average	Max DD	Closed DD	Grade
S47	$31,253	1.30	161	44.3%	$19.96	$3,610	$3,568	C
S47 v2	$24,586	1.49	81	49.7%	$31.32	$1,739	$1,618	A

MultiCharts backtests: S47 2017-01-25 → 2026-10-16, 1566 trades; S47 v2 2017-01-25 → 2026-10-05, 785 trades; costs included, 1 MNQ (×10 = 1 NQ). Max DD counts closed equity plus each trade's worst open loss.

Equity path and drawdown



The trade story

The idea: Europe sets the direction first. S47 measures how far NQ travels around the London cash open — 15 minutes before to 30 minutes after 08:00 London — in units of the overnight range. If that push is big enough it waits out New York's first hour, then trades the same way from 09:31 CT with a stop sized to the overnight range, and closes at 15:00 CT. **v2** also requires New York's own first hour (08:30 → 09:30 CT) to point the same way: the European push confirmed by the US open, in the same market.

Rules

Volatility unit	ATR_ON = average overnight range (17:01 → 08:30 CT) of the previous 20 trading days, at least 15 usable.
London signal	s = (close at London 08:30 − close at London 07:45) ÷ ATR_ON — normally 02:30 and 01:45 CT. Long if s > +0.0625, short if s < −0.0625.
v2 only	**Trade only if 09:30 CT close − last close at or before 08:30 CT has the same sign as s.**
Entry / exits	Market at the 09:31 bar · stop 0.5 × ATR_ON from the bar after entry · exit at the 15:00 CT close · no target · one trade a day.
Skipped days	Quarterly roll days, early-close holidays, unusable overnight sessions.

Worst drawdowns (including open-trade loss)

S47

Depth, 1 MNQ	Peak	Low	Recovered	Days
$3,610.25	2024-01-29	2024-10-04	2025-03-04	400
$3,082.25	2025-04-01	2025-08-01	2026-03-20	353
$3,058.25	2022-07-01	2022-10-12	2023-06-26	360

S47 v2

Depth, 1 MNQ	Peak	Low	Recovered	Days
$1,738.75	2022-07-27	2022-10-12	2022-10-21	86
$1,630.25	2025-05-02	2025-08-01	2025-11-06	188
$1,599.25	2024-01-29	2024-07-17	2024-08-07	191

Now: S47 −$1,541.00 and S47 v2 −$751.50 from their peaks.

By year, 1 MNQ

Year	S47 trades	S47 net	v2 trades	v2 net
2017	149	−$523.50	75	$171.00
2018	165	$4,391.00	78	$1,887.00
2019	145	$223.50	69	$800.50
2020	162	$4,626.50	74	$2,884.00
2021	159	$4,024.00	77	$2,684.50
2022	180	$7,883.00	89	$5,762.00
2023	174	$2,202.50	104	$1,932.00
2024	160	−$2,106.00	84	$505.00
2025	152	$4,219.00	77	$5,845.50
2026	120	$6,313.00	58	$2,115.00

2026 to the end of each backtest.

Why it should work

The direction set at the European open carries into the US session once its first hour is out of the way. There is no published effect size for this exact rule.

- **Signal vs random direction** (same days, clock and stop, 2017–2022): a random direction averages PF 1.14; the London signal 1.41 (p 0.003 before correcting for the ≈3,700 grid cells searched).
- **Other index futures**, same frozen settings: ES PF 1.26 (random 95th percentile 1.17), YM 1.19 (1.18).
- **v2**: in development, agreement days made PF 1.64 and were positive every year; disagreement days made PF 1.23 and lost in 2017 and 2019 (a split made after seeing the trades; the per-trade difference has t ≈ 1.6). Out of sample v2 stayed positive in 2024 ($505.00) when plain S47 lost −$2,106.00.

Risks and caveats

- **Regime-dependent:** most development profit came from sell-off periods (2018 H2, 2020 H1, 2022 H1: 76% of net); the calm years 2017 and 2019 were flat to negative for S47.
- **Tail-dependent:** the top 10 trades are 42% of S47's development net.
- **Out of sample:** each rule was frozen before its 2023 → run and each run was made once, but the same 2023 → window was read four times for this signal: a VXN-gated twin, then S47 (2026-10-03), then v2 and a release-day variant (2026-10-08). S47 was kept over the gated twin after both results, and v2 survived while the release-day variant died. The v2 condition was found on the 2017–2022 trades, after S47's 2023 → result was known. The clean test is data after 2026-10-08.
- **Drawdown:** worst so far $3,610 for S47 against $1,739 for v2 (1 MNQ, incl. open loss).

Robustness grade

Split at 2023-01-01, 1 MNQ	S47 C	S47 v2 A
Trades in / out of sample	960 / 606	462 / 323
Expected value per trade, in → out	$21.48 → $17.54	$30.71 → $32.19
Kept out of sample · P(edge ≤ 0)	82% · 0.0379	105% · 0.0101
Out-of-sample net / max DD (closed)	$10,628 / $3,568	$10,398 / $1,508
Return ÷ drawdown, out of sample	2.98	6.89
Monte-Carlo max DD, median / 75th / 95th / 99th	$3,711 / $4,527 / $6,184 / $7,831	$2,409 / $2,930 / $4,018 / $5,046

Expected value from a 10,000-draw bootstrap; drawdowns from a 20,000-path Monte Carlo of the trade sequence (closed equity). Risk unit = in-sample average loss ($93.40 and $98.63 per MNQ). Grade scale S–D combines edge kept out of sample, confidence it is above zero, and return ÷ drawdown. Each rule was frozen before its single 2023 → run; the window itself was read four times for this signal (see Risks), so the out-of-sample column is not a first look.

Development and out-of-sample runs, 1 MNQ

	S47 · development	S47 · out of sample	v2 · in sample	v2 · out of sample
Window	2017 → 2022	2023-01-25 → 2026-09-01	2017-01-25 → 2022-12-30	2023-01-25 → 2026-10-01
Source	Python	MultiCharts	MultiCharts	MultiCharts
Trades	970 (≈162 / yr)	581	462 (≈77 / yr)	319
Net	$20,533	$11,296	$14,189	$10,840.50
Profit factor	1.411	1.22	1.64	1.40
Win rate	44.1%	44.06%	51.08%	47.96%
Average trade	$21.17	$19.44	$30.71	$33.98
Max drawdown	$2,938 (closed)	$3,636	$1,888.50	$2,559

Runs at $15 per NQ round trip, shown per MNQ (÷10). Max drawdown is MultiCharts' max strategy drawdown except where marked closed. The v2 in-sample run matched the Python emulation to the dollar.

Tried and rejected — do not repeat

When	Tried	Result (1 MNQ, 2017–2022 unless noted)	Verdict
2026-10-08	Stop 0.3 / 0.4 / 0.6 × ATR_ON instead of 0.5	0.4: net $20,259.50 vs $20,533, PF 1.451 vs 1.411, closed DD $2,202.50 vs $2,938; PF smooth from 0.3 to 0.6	Not adopted — rules stay frozen
2026-10-08	ATR regime filter (ATR20 ÷ ATR100: ≥ 0.8, 0.6–1.2, ≤ 1.5, 0.8–1.5)	Misses the worst drawdown (it sits in a normal-ATR period); an upper cap cuts profit from volatility-expansion days	Rejected
2026-10-08	Entry at 10:00 instead of 09:31 (MultiCharts)	Net −$3,123.50 lower (−14%), closed drawdown 26% lower	Not adopted
2026-10-08	Weekday filter	Differences are noise (F-test p 0.72)	Rejected
2026-10-08	Clock-hour analysis	No losing hour; 56% of stop-outs come before 11:00	No filter
2026-10-08	Quiet-day measures: opening-hour range, overnight range, opening volume, ADX(14) on 15-min and daily bars	None orders the trades	Rejected
2026-10-08	Release days only (NFP, CPI, PCE, GDP, ISM, FOMC)	In sample PF 1.90; out of sample 2023-01-27 → 2026-10-01: PF 1.03, $417, closed DD $5,185	Dead
2026-10-08	London and New York agree	In sample PF 1.41 → 1.64, every year positive; out of sample PF 1.40	Adopted as v2
2026-09-30	Same signal with a VXN gate or an ES-agreement gate (C04)	Out of sample (VXN gate): 363 trades, $7,390, PF 1.20 vs plain S47's $11,296, PF 1.22	S47 kept over it, chosen after both 2023 → results

2017–2022 screens use the Python reference (970/970 trades identical to an emulation of the PowerLanguage file) or the native 2017–2022 trade list. The stop, regime, entry-time, weekday, hour and quiet-day screens did not use 2023 →; the release-day, agreement and gate rows were decided on their 2023 → runs.

Recent window (S47 v2)

Backtest 2023-01-25 → 2026-10-05: 87 trades a year · net $10,722.0 · PF 1.39 · average $33.40 · max drawdown $1,630.25 (closed $1,508.00) · last 3 months $901.0.

After 2024 — does the regime still show up?

For v2, yes: $505.00 in 2024 (weak but positive), $5,845.50 in 2025 and $2,115.00 in 2026 to 2026-10-05. Plain S47 lost −$2,106.00 in 2024, then made $4,219.00 in 2025 and $6,313.00 in 2026 to 2026-10-16 (1 MNQ). The calm-market weakness seen in 2017 and 2019 is the thing to watch.

Kill test — what would prove it wrong

Watch	S47	S47 v2	Warning sign
Drawdown	Worst $3,610; Monte-Carlo 75th $4,527	Worst $1,739; Monte-Carlo 75th $2,930	Beyond the 75th percentile.
Win rate vs the win rate it needs to make money	44.3% vs 38.0% needed	49.7% vs 39.9% needed	Win rate sliding toward the needed level.
Trade count	≈ 161 a year	≈ 81 a year	Materially different: setup drift.
Signal vs random direction	Clearly above a random direction on the same days (PF ≈ 1.14 on NQ)		Falls to the random level.
v2 agreement	Agreement days beat disagreement days		No difference: v2 is just S47 with fewer trades.

S47: long 808 $14,751, short 758 $16,502, largest loss −$477.00, positive months 66/118. S47 v2: long 410 $10,195, short 375 $14,392, largest loss −$460.00, positive months 79/118. Source: MultiCharts backtests; rules from the strategy code. Backtests, not account results. Prepared 2026-10-22.